

Smart MCQ Solver Challenge — Milestone 3 Analysis

Milestone 3 Technical Evaluation Report: Dense Vector Search, Cross-Encoder Reranking, and Zero-Shot RAG Augmentation

Competition:	Smart MCQ Solver Challenge
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Core Scope:	Retrieval-Augmented Generation (RAG), FAISS Indexing, Cross-Encoder Reranking, Adversarial Context Audits

Objective Overview: This report details the implementation and evaluation of a Retrieval-Augmented Generation (RAG) framework for multiple-choice question answering. We constructed a 2,000-document knowledge base from the ground-truth target answers in train.csv, indexed them using a dense FAISS vector index (all-MiniLM-L6-v2), and implemented a two-stage retrieve-and-rerank pipeline with a Cross-Encoder (ms-marco-MiniLM-L-6-v2) before passing augmented prompts to a zero-shot classifier (facebook/bart-large-mnli). We systematically audited retrieval recall, reranking promotion, context token budgets, and adversarial context degradation.

1. Zero-Shot Baseline & Bi-Encoder FAISS Vector Search

To evaluate the baseline classification capability without external context, we ran zero-shot classification on Question Row ID 150 using facebook/bart-large-mnli. Subsequently, we built a FAISS vector index using sentence-transformers/all-MiniLM-L6-v2 over all 2,000 correct option texts in the dataset to perform dense bi-encoder semantic retrieval.

Evaluated Metric / Pipeline Step	Verified Technical Value	Analytical Observation
Baseline Zero-Shot P(correct option)	0.384 (38.4%)	Unaugmented BART model shows uncertainty across choices on Row 150.
Knowledge Base Size	2,000 documents	One document per training row representing correct option text.
FAISS Embedding Dimension	384 dimensions	Dense vector representations via all-MiniLM-L6-v2 using IndexFlatL2.
FAISS Top-10 Retrieval Order (Row 150)	[663, 1701, 1269, 1532, 576, 847, 1693, 1906, 168, 150]	Dense bi-encoder returns candidate document indices based on L2 distance.
FAISS Initial Rank of True Document	Rank 10	The true document (Index 150) was retrieved, but at the tail of top-10 candidates.

2. Two-Stage Cross-Encoder Reranking & Context Length Audit

While bi-encoder models allow fast approximate retrieval over large vector spaces, they embed questions and documents independently. A **Cross-Encoder** (cross-encoder/ms-marco-MiniLM-L-6-v2) processes question-document pairs simultaneously, allowing full self-attention across tokens to refine candidate scoring.

Key Reranking Findings:

- **Cross-Encoder Reranked Order (Row 150):** [150, 1906, 847, 1693, 1269, 1532, 168, 576, 1701, 663]
- **True Document Rank Promotion:** Promoted from **Rank 10** (bi-encoder search) to **Rank 1** after Cross-Encoder reranking.
- **Context Token Audit (Row 42):** Concatenating top-5 retrieved documents into a RAG prompt format (Context: <docs> Question: <prompt>) and tokenizing with bert-base-uncased yields exactly **216 tokens**, comfortably fitting standard transformer context limits (512 tokens).

3. RAG Context Augmentation & Adversarial Stress-Testing

To measure the impact of retrieved context on model decisions, we conducted an A/B experiment comparing ground-truth context augmentation against an adversarial distractor attack (injecting an unrelated document, KB Index 999).

Evaluation Scenario	Context Description	P(Correct Option)	Impact / Performance Delta
Zero-Shot Baseline (No Context)	Raw question prompt without context	0.384	Baseline reference level
True Ground-Truth RAG Context	Concatenated true document (KB Index 150)	0.989 (98.9%)	+0.605 (+157.6% boost in confidence)
Adversarial Distractor RAG Context	Forced unrelated document (KB Index 999)	0.529 (52.9%)	-0.460 degradation from true context

Adversarial Vulnerability Analysis: Forcing a completely unrelated context document (KB Index 999) caused the correct option probability to drop from 0.989 down to 0.529. This empirical result demonstrates the classic *garbage-in, garbage-out* vulnerability of retrieval-augmented systems: LLM classifiers place strong trust in prompt context, making high-precision retrieval and reranking essential.

4. Benchmark Hit Rate & End-to-End Pipeline Performance

We scaled our evaluation across multiple benchmark subsets to measure overall retrieval recall and end-to-end question answering accuracy (MAP@3).

Benchmark Metric	Evaluation Subset	Verified Score	Technical Significance
Top-5 Retrieval Hit Rate	First 100 Training Questions	73.0%	73 out of 100 prompts retrieved the ground-truth document in the top 5 candidates.
Full RAG Pipeline MAP@3	First 20 Evaluation Questions	0.975	Complete Retrieve -> Rerank -> Augment -> Classify pipeline achieved near-perfect top-3 ranking precision.

5. Strategic Takeaways & Architectural Recommendations

- **Two-Stage Pipeline Value:** Bi-encoder FAISS search rapidly reduces the search space to 10 candidates, while Cross-Encoder reranking successfully elevates true documents from tail ranks (Rank 10) to top rank (Rank 1).
- **Context Augmentation Gains:** Providing accurate context boosts model classification probability from 0.384 to 0.989, enabling near-perfect MAP@3 performance (0.975).
- **Mitigating Distractor Risks:** Cross-Encoder reranking acts as a critical filter against adversarial or noisy context documents, preventing model hallucinations and misclassifications.